



TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
NATIONAL COLLEGE OF ENGINEERING

A
MINOR PROJECT REPORT
ON
**“ENSEMBLE LEARNING BASED LANDSLIDE RISK
ASSESSMENT MODEL”**

SUBMITTED BY:

BIKASH DHAMI (NCE080BCT010)
GALIJ SUNUWAR (NCE080BCT015)
NALISHA SHAKYA (NCE080BCT023)
SUMOON BYANJANKAR (NCE080BCT045)

SUBMITTED TO:

DEPARTMENT OF ELECTRONICS & COMPUTER ENGINEERING

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Page of Approval

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DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

The undersigned certifies that they have read and recommended to the Institute of Engineering for acceptance of a project report entitled “**Ensemble Learning Based Landslide Risk Assessment Model**” submitted by **Bikash Dhimi, Galij Sunuwar, Nalisha Shakya**, and **Sumoon Byanjankar** in partial fulfillment of the requirements for the Bachelor’s degree in Electronics, Information and Communication Engineering & Computer Engineering.

.....

Supervisor

Suroj Burlakoti

Head of Department,

Department of Electronics and Computer

Engineering,

National College of Engineering, IOE, TU.

.....

External examiner

Full Name

Designation

Affiliation

Date of approval: August, 2026

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Head
Department of Electronics and Computer Engineering
National College of Engineering
Talchhikhel, Lalitpur.

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Abstract

Landslides remain one of Nepal’s most damaging natural hazards, and Dhading district, with its steep terrain and intense monsoon rainfall, has a severe recurring history but no localized predictive mapping tool. This project develops an ensemble machine learning system for landslide risk assessment across six municipalities of Lower Dhading, delivering both a district-wide susceptibility zonation and an on-demand, location-specific risk query system. A pixel-level, monthly-resolved dataset of approximately 39.8 million rows was assembled from multi-source geospatial data, terrain, vegetation, rainfall, land cover, and infrastructure proximity, against a manually verified historical landslide inventory. Severe class imbalance (1:1216) was addressed through spatial buffer exclusion and grid thinning, and an Optuna-tuned stacking ensemble of four tree-based classifiers was trained using leakage-free, location-grouped out-of-fold predictions and evaluated on a spatially held-out test set to prevent location leakage between training and testing. All base learners exceeded 0.96 ROC-AUC, with the stacked ensemble achieving the highest recall of any configuration (0.872) at a tuned threshold, prioritizing hazard-detection sensitivity while matching the best individual learner’s ranking quality. This recall gain confirmed the base models captured complementary rather than redundant patterns, and landslide rates rose sharply with slope and fell with vegetation density, confirming physical plausibility. Applied district-wide, the model classified 308 zones into four risk bands, identifying 10 zones as High or Very High risk, concentrated along the Prithvi Highway corridor. The study demonstrates that a stacked ensemble can outperform its best individual constituent for hazard-warning sensitivity, delivered as a deployed query system filling a documented localized-mapping gap for Dhading district.

Keywords: *Landslide Susceptibility Mapping, Ensemble Learning, Stacking, Random Forest, XGBoost, LightGBM, CatBoost, Class Imbalance, Geospatial Machine Learning, Dhading*

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List of Abbreviations

Abbreviation	Full Form
ACTE	Advanced Career and Technical Education
AOI	Area of Interest
API	Application Programming Interface
AUC	Area Under Curve
BIPAD	Building Information Platform Against Disaster
CART	Classification and Regression Tree
CHIRPS	Climate Hazards group Infrared Precipitation with Stations
CMIP6	Coupled Model Intercomparison Project Phase 6
CORS	Cross Origin Resource Sharing
CRS	Coordinate Reference System
CSV	Comma Separated Values
DEM	Digital Elevation Model
DHM	Department of Hydrology and Meteorology
DMG	Department of Mines and Geology
DNN	Deep Neural Network
EFB	Exclusive Feature Bundling
GB	GigaByte
GEE	Google Earth Engine
GIS	Geographical Information System
GOSS	Gradient-based One-Side Sampling
LR	Logistic Regression
LSM	Landslide Susceptibility Mapping
LULC	Land Use/Land Cover
ML	Machine Learning
MLP-ANN	Multilayer Perceptron Artificial Neural Network
NASA	National Aeronautics and Space Administration
NB	Naïve Bayes
NDRRMA	National Disaster Risk Reduction and Management Authority

NDVI	Normalized Difference Vegetation Index
OOF	out-of-fold
OSM	OpenStreet Maps
PR	Precision-Recall
QGIS	Quantum Geographic Information System
RF	Random Forest
ROC	Receiver Operating Curve
SPI	Stream Power Index
SSP	Shared Socioeconomic Pathway
STI	Sediment Transport Index
SVM	Support Vector Machine
XGBoost	Extreme Gradient Boosting

1. Introduction

Landslides in Nepal claim hundreds of lives and cause severe economic damage every year, yet many of the country’s most landslide-prone districts lack the predictive mapping tools that would allow authorities to prepare rather than merely respond. This chapter introduces the background, problem statement, objectives, and scope of the project.

1.1 Background

A landslide is the movement of rock, soil, and organic matter down a slope, driven by gravitational force acting on materials that have lost their shear strength. It is among the most devastating natural disasters worldwide, causing extensive damage to vegetation, property, and human life, with frequency expected to rise due to increased urbanization, deforestation, and intensified precipitation from climate change. A Landslide Susceptibility Map (LSM) addresses this by classifying terrain into relative hazard zones, giving planners and authorities a spatial basis for mitigation rather than reactive, post-disaster response. Despite this value, a large fraction of the world’s most landslide-prone terrain remains inadequately mapped, and these gaps do not necessarily correspond to areas of lower hazard. This is starkly illustrated in Nepal, where a comprehensive review of national landslide inventories found that a large share of the country had been surveyed only once in over a decade, with existing records concentrated around specific events and major highway corridors rather than distributed according to actual hazard [1].

Closing such gaps efficiently, without relying on exhaustive field surveys, is increasingly feasible through data-driven susceptibility mapping built on freely available GIS and remote sensing datasets. This scalability has been demonstrated even across some of the most extensive and geologically complex mountainous regions on Earth, where data-driven models have consistently proven capable of identifying the dominant physical controls on landslide occurrence across highly variable terrain [2]. Nepal itself is highly prone to landslides due to its steep terrain, weak rock structure, and heavy monsoon rainfall, compounded by unplanned urbanization and poor development planning, highlighting an urgent need for a structured resilience strategy. This vulnerability has already prompted susceptibility assessments at the local level but such coverage remains confined to only a small administrative fraction of individual districts, such as a single rural municipality within Dhading, rather than offering a comprehensive, district-wide assessment [3].

Where such assessments have been carried out, they have typically relied on individual

machine learning classifiers, including Random Forest, Support Vector Machines, and Artificial Neural Networks, applied both within Nepal and internationally, with literature surveys confirming their broad adoption and growing diversity as landslide risk intensifies globally [4]. Within Dhading district specifically, the work that does exist has largely been limited to field-based landslide inventory documentation rather than predictive susceptibility modelling, cataloguing debris flows, debris slides, and deep-seated creeps without translating this record into a zonation map that could inform mitigation planning [5]. This local gap is compounded by a broader mismatch at the national scale, where susceptibility and population exposure have been found to vary considerably and inconsistently across the country, underscoring the limitations of relying on coarse, national-level susceptibility layers in place of localized, district-specific assessment [6]. At a regional scale closer to the present study area, causative-factor analyses across the Nepal Himalaya have consistently identified slope, geology, and road construction activity as dominant drivers of landslide occurrence, but have continued to rely on classical statistical regression rather than more advanced ensemble techniques, even in comparatively recent work [7].

This methodological gap is not unique to Nepal, and ensemble learning, which combines multiple base classifiers into a single, more robust prediction, has emerged globally as a leading response to it. Comparative studies have repeatedly shown that a well-constructed ensemble can outperform every one of its individual constituent classifiers on the same dataset [8]. The specific ensemble strategy chosen, however, meaningfully affects the resulting performance gain: boosting-based approaches have generally been found to yield stronger and more consistent improvements over standalone baselines than alternatives such as stacking, particularly under rainfall-triggered conditions [9]. This advantage holds even when ensemble techniques are paired with neural network base classifiers rather than simpler statistical or tree-based learners, with every ensemble configuration tested improving on its standalone baseline in such comparisons [10].

Dhading district, the focus of the study, reflects this exact combination of problems: a severe history of recurring landslides at the ward level, only partial and non-predictive susceptibility coverage, and no application of ensemble learning despite its demonstrated advantages elsewhere. This absence of baseline mapping, combined with the well-documented data gaps affecting landslide-prone regions and the consistent performance gains offered by ensemble methods, directly motivates the ensemble learning-based approach adopted in the study to combine multiple GIS-derived conditioning factors and generate a reliable susceptibility map for the region.

1.2 Problem Statement

Landslide inventory documentation in Nepal has increased significantly over the past decade, yet disaster management agencies such as the DHM, DMG, and NDRRMA continue to rely mostly on reactive strategies, focusing on mapping damage after an event or issuing broad rainfall threshold alerts rather than predicting risk beforehand. This gap persists largely because current methods fail to combine multiple high-resolution geospatial factors, such as slope angle, vegetation indices, elevation, land cover, and road proximity, into a single, unified predictive framework, leaving no proactive landslide risk prediction system in place for vulnerable regions. As a result, districts such as Dhading face recurring monsoon disasters without a data-driven tool to prepare for localized risk, with communities in lower-lying areas remaining exposed until after a disaster has already occurred. The existing limited works for the district has been restricted to field-based inventory documentation rather than a predictive susceptibility framework capable of informing mitigation planning. This absence of a unified, localized framework also leaves local leaders and response teams without location-specific decision-making support during the monsoon season, while ongoing road infrastructure expansion in hilly terrain continues to destabilize slopes, a factor repeatedly identified as one of the dominant causes of landslide occurrence across the wider Nepal Himalaya, yet one that remains largely unaccounted for in existing localized risk assessments.

1.3 Objectives

- To train and validate an ensemble machine learning model on multi-source geospatial conditioning factors for landslide susceptibility within the study area.
- To generate a ward-level landslide susceptibility zonation map using the trained model, and enable on-demand landslide risk querying for a specific location.

1.4 Scope

This project focuses on the lower Dhading region of Dhading district, Nepal, comprising six rural municipalities: Benighat Rorang, Siddhalek, Gajuri, Galchhi, Thakre, and Dhunibensi, analyzing its unique terrain, road networks, and historical landslide inventory. The study concentrates entirely on software design, machine learning model training, and geospatial data analysis. It does not involve deploying physical hardware sensors, collecting raw geotechnical soil samples through field testing, or constructing structural engineering mitigation measures.

The system and its generated susceptibility maps are particularly relevant for:

- Local Government and Ward Authorities: Local leaders needing ward-level predictive tools for resource allocation and safety decisions before monsoon season.
- Disaster Response and Mitigation Teams: Emergency agencies requiring predictive spatial intelligence for proactive hazard zoning over reactive relief.
- Infrastructure and Road Planning Bodies: Department of Roads and local engineers using susceptibility maps to minimize landslide triggers during road expansion and hilly terrain construction.
- Vulnerable Rural Communities: Residents in high-slope or landslide-prone areas of Dhading who benefit from long-term, data-driven risk planning.

2. Literature Review

This chapter reviews existing work on landslide susceptibility mapping and presents the theoretical foundations underlying the models and methods used in this project.

2.1 Related Work

Across the entire Qinghai-Tibetan Plateau, Sajadi et al. assessed landslide susceptibility using five ML algorithms (DNN, LR, NB, RF, SVM) on 671 events and 13 conditioning factors. RF performed best, with rainfall, elevation, and lithology identified as the top controlling factors [2]. This growing preference for data-driven methods over classical statistical models is corroborated at a much larger scale by Reichenbach et al., whose review of 565 statistically-based susceptibility studies published between 1983 and 2016 found a clear and increasing shift toward machine learning approaches in more recent literature [11]. A bibliometric knowledge-mapping analysis by Yong et al., covering susceptibility research indexed in the Web of Science between 1991 and 2020, similarly identified ensemble and intelligent models as one of the field’s key emerging directions [12].

The scale of Nepal’s data gap has been documented directly by Harvey et al., who reviewed 117 landslide inventories for Nepal between 2010 and 2021 and found that 40% of the country had been mapped only once, with mapping efforts concentrated around high-profile events such as the 2015 Gorkha earthquake and major highways rather than distributed according to actual hazard [1]. A complementary national-scale picture emerges from Kincey et al., who modeled susceptibility across the whole of Nepal and combined it with population and building data to assess human exposure, validating their model against landslide incidents and fatalities recorded on the Government of Nepal’s BIPAD disaster information portal, finding that although susceptibility was highest in the High Himalaya, actual population exposure peaked in the Middle Hills, with both estimates shown to be highly sensitive to the limited length of existing landslide records [6]. Following a similar approach, the BIPAD portal was initially explored as a potential source of ward-level landslide occurrence data for Dhading district in the study. However, the available records for the district were found to be too sparse for reliable model training, prompting a shift toward deriving landslide inventory data directly from satellite imagery instead.

Closer to the present study area, K.C. et al. carried out field-based landslide inventory mapping across the Ramche-Jharlang area spanning parts of Dhading, Rasuwa, and Nuwakot districts, documenting landslide type, geometry, and triggers through direct field surveys

and community interviews, and identifying weak phyllite lithology, road-cut excavation, and monsoon precipitation as the dominant causes [5]. Within Dhading district itself, Dhungana et al. applied the Analytical Hierarchy Process for landslide susceptibility and risk mapping in Benighat Rural Municipality, integrating fourteen conditioning factors alongside social vulnerability indicators to produce a risk map intended to guide resilient local planning [3]. At a broader regional scale, Dahal identified slope, aspect, geology, and road construction as the dominant causative parameters governing landslide activity across roughly 4,000 km² of central Nepal using logistic regression [7], while earlier work by Dahal et al. applied weights-of-evidence modelling to rainfall-induced landslides in the hills bordering the Kathmandu Valley following the damaging 2002 monsoon [13]. Even recent work by Thapa et al. in Nepal’s ecologically fragile Chure region continued to rely on classical Poisson regression rather than ensemble methods, identifying geology, lineament density, slope, and rainfall as significant predictors [14].

A broader methodological survey by Ado et al. examined machine learning models used for landslide susceptibility mapping, finding that many exceeded an AUC of 0.90 and highlighting the growing prevalence of hybrid, ensemble, and deep learning approaches [4]. In Baglung district, Chalise et al. compared Random Forest and CART for susceptibility mapping using ten causative factors, with RF achieving a higher success rate and slightly better overall accuracy than CART [15]. Kansakar and Dahal produced susceptibility maps for Lamjung district using the XGBoost algorithm under both baseline and projected future climate scenarios, finding a marked increase in high-risk area under future rainfall projections compared to historical baselines [16]. Pradhan and Kim applied a Multilayer Perceptron Artificial Neural Network in the Phewa catchment of western Nepal, achieving a success rate of 82.1% and a prediction accuracy of 91.4% using an inventory of 88 landslide locations [17].

Outside Nepal, Marjanović et al. compared Support Vector Machines, Decision Trees, and Logistic Regression for the Fruška Gora Mountain region of Serbia, finding that SVM consistently outperformed the other models and even proved more spatially reliable than the traditional Analytical Hierarchy Process [18]. Building on such single-classifier results, Kalantar et al. compared four individual algorithms against an ensemble combining all of them, finding that the ensemble model outperformed every individual constituent classifier across all evaluation metrics [8]. The choice of ensemble strategy was shown to matter considerably by Dou et al., who combined an SVM base classifier separately with bagging, boosting, and stacking for a watershed in Kyushu Island, Japan, finding that boosting substantially outperformed both the standalone model and the stacking variant [9]. A similar advantage was demonstrated by Pham et al. in the Himalayan region of India, where six dif-

ferent ensemble techniques paired with an MLP Neural Network base classifier all improved on the standalone baseline, with MultiBoost achieving the strongest overall performance [10].

2.2 Related Theory

2.2.1 Independent Input Variables

2.2.1.1 Normalized Difference Vegetation Index (NDVI)

NDVI is a widely used satellite-based indicator of vegetation health. It measures the difference between near-infrared (which healthy vegetation reflects) and red light (which healthy vegetation absorbs). A lower NDVI value indicates dry, low vegetation, which increases the probability of a landslide. Higher NDVI value indicates dense vegetation, which includes root binding with soil, reduces soil erosion, improves slope stability, and reduces the probability of landslides.

2.2.1.2 Rainfall

Rainfall is one of the primary hydrological triggers of landslides, as intensified precipitation increases pore water pressure and reduces the shear strength of slope material. Higher cumulative and short-duration rainfall intensity raises soil saturation and destabilizes slopes, particularly during the monsoon season, while prolonged dry periods allow soil cohesion to recover.

2.2.1.3 Distance to Roads

Distance to roads captures the influence of road construction and cutting on slope stability. Areas closer to roads face a higher landslide risk due to slope-cutting, altered drainage, and loss of toe support during construction. Risk decreases as distance from road infrastructure increases.

2.2.1.4 Distance to Rivers

Distance to rivers reflects the influence of fluvial erosion and saturation on nearby slopes. Slopes closer to rivers are more susceptible to landslides due to continuous toe erosion and elevated soil moisture, while slopes farther from river channels are comparatively more stable.

2.2.1.5 Land Use/Land Cover (LULC)

LULC classifies the surface cover of a region, such as forest, cropland, built-up area, or bare land, and reflects both natural vegetation cover and human land-use decisions. Bare land, cropland, and built-up areas are more prone to landslides due to reduced root reinforcement and soil disturbance, while dense forest cover improves slope stability.

2.2.1.6 Soil Type

Soil type reflects the subsurface material and its engineering behaviour. Clay-rich soils retain water and lose shear strength when saturated, while sandy or gravelly soils drain quickly

but offer limited cohesion. Loosely packed soils on steep terrain are particularly prone to failure during intense rainfall, making soil type an important conditioning factor for landslide susceptibility.

2.2.1.7 Topography (Slope, Aspect, Elevation)

Topographic features are characteristics of the land surface that define its shape. They play a significant role in soil movement and behavior. The risk of landslides is higher in steeper slope areas. South-facing slopes (in the Northern Hemisphere) receive more sunlight, resulting in drier vegetation. Higher elevations have steeper and thinner soil, resulting in a high risk of landslides.

2.2.2 Prediction Algorithms

2.2.2.1 Random Forest (RF)

RF is an ensemble learning method of individual binary decision trees to produce higher predictions with broad applications in regression, classification, and feature selection. RF predicts by taking the average of the votes of different decision trees. Increasing the number of trees reduces overfitting at the expense of computational complexity.

Working:

- Randomly samples subsets of training data and features to construct each decision tree.
- Trains multiple decision trees independently to form the “forest.”
- Aggregates individual tree predictions through voting (classification) or averaging (regression) to produce a final, more robust output.

Formula:

$$\hat{y} = \frac{1}{N} \sum_{i=1}^N T_i(x) \quad (2.1)$$

Where:

- $\hat{y}$ = Predicted value for sample
- $T_i(x)$ = Prediction from the i^{th} decision tree
- N = Number of trees in the forest
- x = Input features (e.g. slope, elevation, etc.)

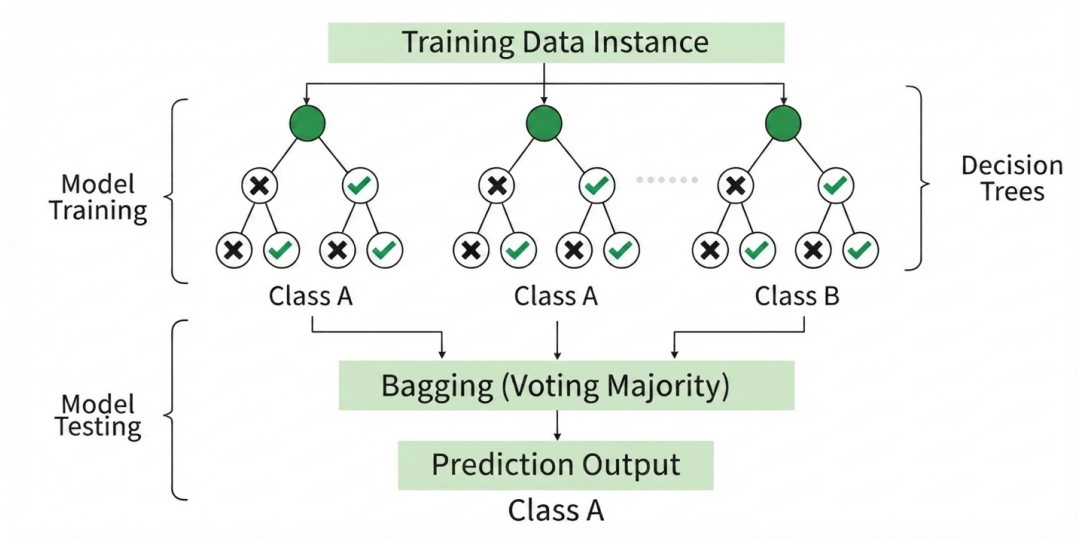


Figure 2.1: Random Forest Architecture

Source: *GeeksforGeeks*

2.2.2.2 Extreme Gradient Boosting (XGBoost)

XGBoost is a gradient boosting algorithm that builds decision trees sequentially, where each new tree is trained to correct the errors of the previous ones. It is widely used for landslide susceptibility prediction by modeling complex relationships among conditioning factors such as slope, rainfall, and NDVI. By iteratively minimizing residual errors, XGBoost improves predictive accuracy and provides feature importance scores to identify the most influential factors. This enables effective mapping of areas with higher landslide risk.

Working:

- Initializes with a base model and iteratively adds trees to minimize residual errors.
- Applies explicit regularization on tree complexity (number of leaves and leaf weights) to reduce overfitting and handles missing values internally.
- Utilizes parallel, block-based processing to enhance computational efficiency on large datasets.

Formula:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i), \quad f_k \in \mathcal{F} \quad (2.2)$$

Where:

- $\hat{y}_i$ = Predicted value for sample i

- K = Number of trees
- f_k = Function representing the k^{th} tree
- x_i = Input features for sample i

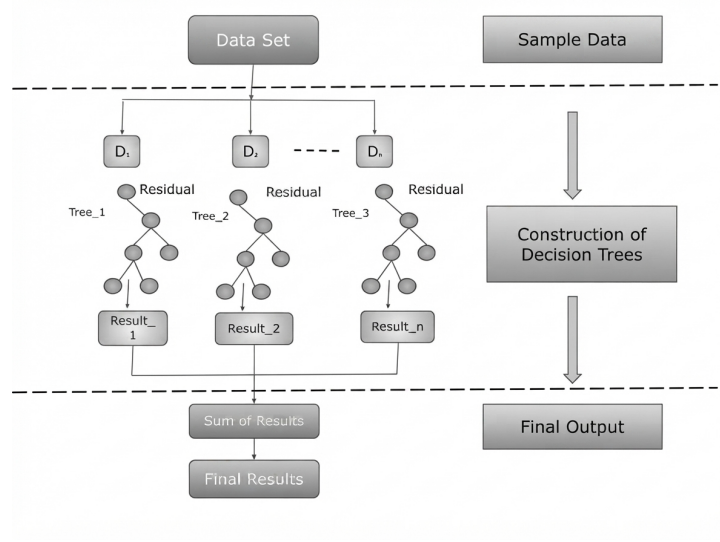


Figure 2.2: Extreme Gradient Boosting (XGBoost) Architecture

Source: TutorialsPoint

2.2.2.3 Light Gradient Boosting Machine (LightGBM)

LightGBM (Light Gradient Boosting Machine) is a high-performance gradient boosting framework designed for speed and efficiency. Like XGBoost, it supports both classification and regression tasks, but it differs in the way trees are constructed. LightGBM grows trees leaf-wise rather than level-wise, allowing it to achieve higher accuracy while remaining computationally efficient, especially when working with large datasets. Its optimized design makes it particularly suitable for large-scale landslide susceptibility modeling.

Working:

- Expands trees by splitting the leaf with the highest information gain (leaf-wise), rather than growing all leaves at a given depth as XGBoost does.
- Uses Gradient-based One-Side Sampling (GOSS) to prioritize poorly predicted samples and Exclusive Feature Bundling (EFB) to merge sparse features, cutting training time.
- Uses histogram-based binning of continuous features to accelerate split-point search on large datasets.

Formula:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i), \quad f_k \in \mathcal{F} \quad (2.3)$$

Where:

- $\hat{y}_i$ = Predicted value for sample i
- K = Number of trees
- f_k = Function representing the k^{th} tree
- x_i = Input features for sample i

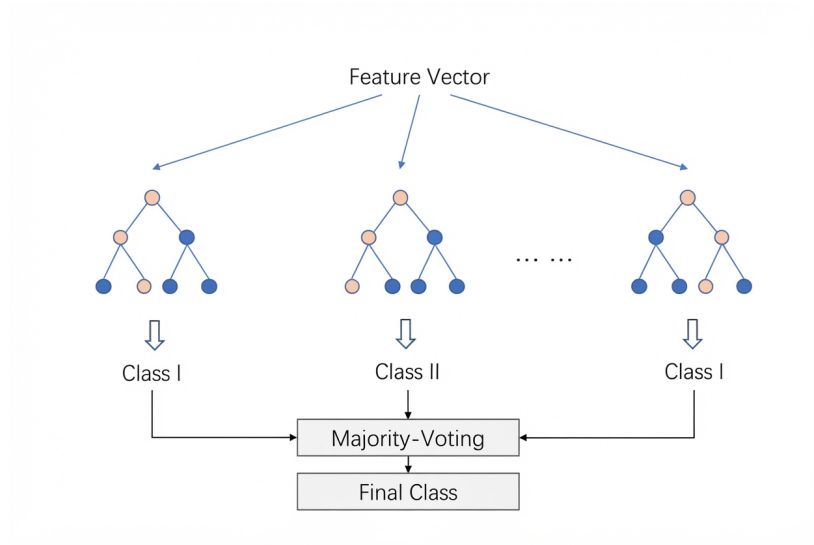


Figure 2.3: Light Gradient Boosting (LightGBM) Architecture

Source: ACTE Technologies

2.2.2.4 Categorical Boosting (CatBoost)

CatBoost (Categorical Boosting) is a gradient boosting algorithm specifically optimized for handling categorical data, employing ordered boosting to reduce overfitting. CatBoost improves predictive accuracy by effectively encoding categorical features and reducing bias in the boosting process.

Working:

- Encodes categorical features (e.g. geology, land cover class) using ordered target statistics rather than one-hot encoding.
- Uses random permutations of the training data to prevent prediction shift in the boosting process.

- Sequentially minimizes the loss function using symmetric (balanced) trees while addressing class imbalances.

Formula:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i), \quad f_k \in \mathcal{F} \quad (2.4)$$

Where:

- $\hat{y}_i$ = Predicted value for sample i
- K = Number of trees
- f_k = Function representing the k^{th} tree
- x_i = Input features for sample i

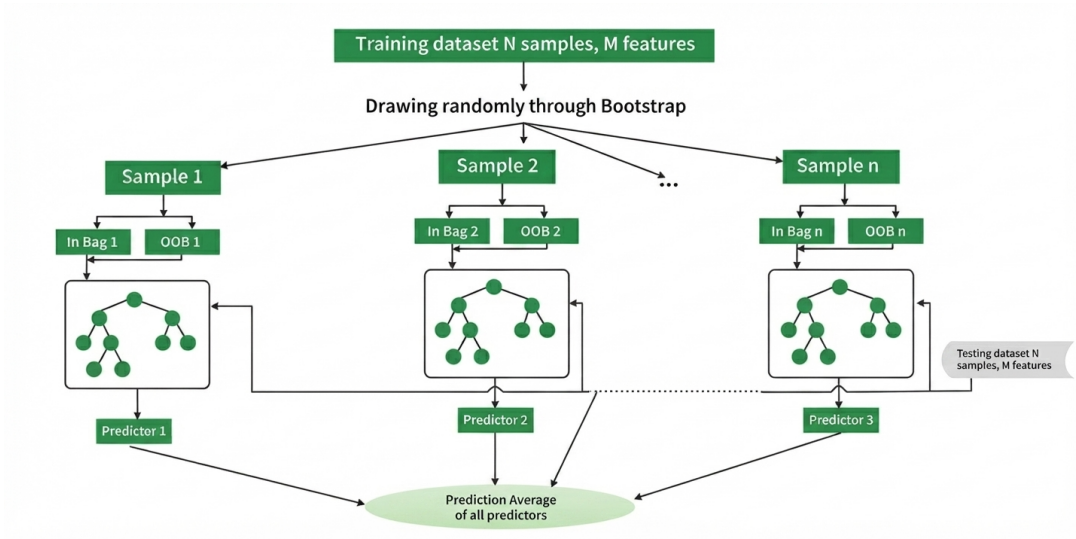


Figure 2.4: Categorical Boosting (CatBoost) Architecture

Source: *GeeksforGeeks*

3. Methodology

This project builds a geospatial machine learning system to assess landslide susceptibility and provide on-demand risk prediction using an ensemble model. Data comes in through Google Earth Engine for most of the variables and a free weather API for live rainfall, passes through the machine learning pipeline, and is delivered via a web interface that a public user and officials can query at any time.

3.1 Location Selection

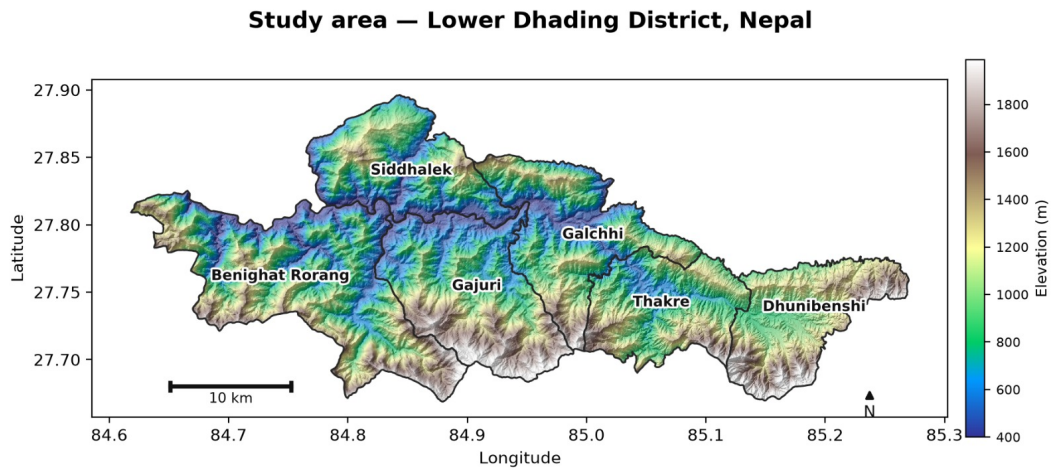


Figure 3.1: Study area: the six municipalities of Lower Dhading, traversed by the Prithvi Highway.

The study covers six municipalities in Lower Dhading region in Nepal as the Prithvi Highway connecting the western region to central region of Nepal goes through the dangerous terrain. The selected municipalities being Benighat Rorang, Dhunibensi, Galchhi, Gajuri, Siddhalek, and Thakre.

Tools Used: QGIS, Python

We used QGIS and Python to first crop out the lower dhading from the whole Nepal geojson map. Then, we rastarized and converted it into tiff file for pixel conversion of 30m resolution. This step was important because it allowed us to extract newer data pixel-wise.

3.2 Data Collection

A major portion was dedicated to the collection, preprocessing and integration of multi-source geospatial datasets required for the landslide susceptibility modeling in the lower Dhading region. We used Google Earth Engine (GEE) and pulled most of the variables using automated Python query script, we used OpenStreetMap (OSM) for one time extraction of distance to road and distance to river, and finally manually extracted positive label via news sources, and even Google Earth Pro satellite.

Table 3.1: Datasets used for landslide susceptibility dataset generation.

S.N.	Dataset	Satellite/Source	Revisit
1	NASADEM	NASADEM	Static
2	NDVI	Sentinel-2A/B	Monthly
3	CHIRPS Daily Precipitation	CHIRPS	Daily
4	Land Use/Land Cover (LULC)	Sentinel-2 + Dynamic World V1	Yearly
5	Soil Class	OpenLandMap	Static
6	Road Network	OpenStreetMap (OSM)	Static
7	River/Waterway Network	OpenStreetMap (OSM)	Static

3.2.1 Historical Landslide Data

Positive landslide observations were identified through manual visual interpretation of high-resolution Google Earth imagery. Historical imagery was examined to detect landslide scars, and the approximate occurrence date of each event was determined by comparing imagery from different acquisition dates. Where available, historical news reports and documented records of known landslide events were used to further constrain the event timing. In addition, significant changes in the Normalized Difference Vegetation Index (NDVI), computed using a Google Earth Engine script, were used as supporting evidence to verify the occurrence period of landslides.

Each verified landslide was recorded with its geographic coordinates and estimated occurrence month, forming the positive class of the historical landslide inventory used for model development. For most landslides, a single event month could be confidently identified and assigned as the positive label. In cases where the exact timing could not be determined with sufficient certainty, positive labels were assigned to two adjacent candidate months to account for temporal uncertainty while preserving the spatial location of the event.

3.3 Data Integration

3.3.1 AOI Masking

All static and periodic rasters were clipped to the lower Dhading study area polygon. Masking is applied by value rather than by shape: rasters remain rectangular, but pixel values outside the AOI boundary are set to each raster’s nodata sentinel. This nodata convention is what ultimately filters out-of-AOI pixels from the final training table at the flattening step, rather than any change in raster dimensions.

3.3.2 Co-registration

Every AOI-masked variable was resampled onto a common 30 m reference grid, anchored to the masked NASADEM elevation raster (matching CRS, transform, and pixel dimensions). Continuous variables (slope, aspect, NDVI) were resampled using bilinear interpolation, categorical variables (LULC) were resampled using nearest-neighbour interpolation to preserve class integrity. Distance-to-road and distance-to-river rasters were passed through directly where already aligned to the reference grid.

Since CHIRPS rainfall data is natively available only at approximately 5.5 km resolution, a direct coordinate join against the 30 m grid would produce almost entirely null values. Each rainfall feature was therefore first rasterized onto its native coarse grid, then bilinear-resampled onto the 30 m reference grid. This means rainfall values in the final table are spatially interpolated within each coarse CHIRPS cell rather than independently measured at 30 m resolution.

3.3.3 Master Table Construction

The final table was built around the vegetation (NDVI) raster as the finest-grained backbone. NDVI rows with nodata values (including all out-of-AOI pixels) were dropped first, establishing the AOI boundary for the table. Rainfall features were joined onto the remaining rows at the same (longitude, latitude, period) cadence. LULC was joined at (longitude, latitude, year) and broadcast across that year’s periods. Static terrain, distance, and soil type variables were joined at (longitude, latitude) and broadcast across every row regardless of period. The complete dataset comprises roughly 39.8 million pixel-period rows across approximately 1.9 million unique locations.

Table 3.2: Final Training Table Schema

Column(s)	Type	Cadence
ndvi	float	Periodic (per month)
lulc	categorical (int code)	Annual, broadcast to periods
rainfall	float	Periodic, bilinear-interpolated
elevation, slope, aspect_sin, aspect_cos, dist_road, dist_river, soil_type	float / categorical	Static
month_sin, month_cos	float	Dynamic (temporal encoding)
is_landslide	binary (0/1)	Target variable

3.3.4 Data Preprocessing

Once the raw dataset was built, the data had to be prepared before it could be used to train the models. Preprocessing ensures that data is clean, formatted and transformed into a usable state. This section explains how the missing values, the aspect angle, and the feature scales were handled, and how the data was split.

Handling the aspect angle: Aspect (the direction a slope faces) is a circular quantity: an angle of 0° and 360° are actually the same direction. If the raw angle is fed to a model as a plain number, the model would wrongly treat 0° and 360° as far apart and 359° and 1° as far apart, even though they are adjacent directions. To fix this, aspect was converted into two features using the sine and cosine of the angle:

$$aspect_sin = \sin(aspect), \quad aspect_cos = \cos(aspect)$$

Handling missing values: The two types of features were handled differently. Numeric features (NDVI, Rainfall, Elevation, Slope, Distance_to_Road, Distance_to_River, aspect_sin, aspect_cos) had missing values filled with the column median. The median was chosen over the mean because it is not affected by the extreme values, which are common in data like rainfall and elevation. Categorical features (Soil_Type, LULC_name) had missing values filled with the most frequent value (mode) in the column.

Scaling numeric features: The numeric features are on very different scales: NDVI is close to zero, while elevation is measured in thousands of metres. To keep any one feature from

dominating the others, all numeric features were standardized to have zero mean ($\mu = 0$) and unit variance ($\sigma^2 = 1$) using a standard scaler.

Encoding categorical features: The categorical features, soil type and land-use/land-cover class, were encoded using one-hot encoding, where each category becomes its own binary column. There are 3 soil types and 9 land-cover classes in the data. New or unseen categories that might appear in live data are simply ignored rather than causing an error.

All of these steps were combined into a single scikit-learn pipeline, so the exact same preprocessing can be applied to new data later without any manual rework.

Splitting the data by location: Every location in the dataset is sampled multiple times across different monsoon months and years. A normal random split could therefore put the same location into both the training and test sets, which would make the model look better than it really is, because it would be tested on places it has already seen. To prevent this, each row was given a group ID made from its latitude and longitude, and the data was split by location instead of by row: all rows of one location stay entirely on one side of the split. An 80/20 split was used with a fixed random seed (42) for reproducibility.

The preprocessing pipeline was fitted on the training data only, and then used to transform both the training and test sets. If it had been fitted on the full dataset, the median values and scaling parameters would have leaked test-set information into training. Fitting on the training data only keeps the test set completely unseen until evaluation.

3.3.5 Model Training

Handling class imbalance: Only about 5.76% of the samples in the dataset are landslides. A model that simply predicted “no landslide” for everything would therefore appear highly accurate without learning anything useful. Two measures were taken to deal with this. First, each model was configured to pay more attention to the landslide class: Random Forest and LightGBM used class weighting `class_weight`, and CatBoost used auto class weights. Second, per-model decision thresholds were used, since the default threshold of 0.5 is often not the best choice for imbalanced data.

Tuning the hyperparameters: The hyperparameters of all four models were tuned using Optuna over 20 trials. The tuning used the same location-based splitting to avoid leakage, and the models were compared using PR-AUC, which suits imbalanced problems better than plain accuracy. The final hyperparameters used for training are shown in Table 3.3.

Stacking the models: The final model is a stacking ensemble of the four base models: Random Forest, XGBoost, CatBoost, and LightGBM. Each base model predicts a landslide probability, and a logistic regression “meta-learner” learns the best way to combine those four probabilities into one final answer.

Table 3.3: Hyperparameters of the four base models.

Model	Key Parameters
Random Forest	300 trees, max depth 14, min 5 samples per leaf
XGBoost	250 trees, max depth 10, learning rate 0.12, with regularization
CatBoost	500 iterations, depth 6, learning rate 0.05
LightGBM	250 trees, 122 leaves, learning rate 0.09

A crucial aspect of the stacking procedure was maintaining a leakage-free training process for the meta-learner. Predictions generated from base models trained on the same samples could introduce information leakage. Hence, out-of-fold (OOF) predictions were implemented to provide unbiased inputs to the meta-learner:

1. The training data was divided into 5 folds, with the splits grouped by location.
2. For each fold, the four base models were trained on the other four folds and then used to predict the held-out fold.
3. This ensured that every training row received predictions from models that had not previously seen that row.
4. These OOF predictions were then used as the input features for training the logistic regression meta-learner.
5. Afterward, the four base models were retrained using the entire training dataset. These models, together with the already-trained meta-learner, formed the final stacking model used to make predictions on new data.

Decision thresholds: Models normally classify a sample as positive when its probability is above 0.5, but with imbalanced data that default is often not the best choice. A threshold was therefore picked for each model to balance precision (avoid too many false alarms) and recall (avoid missing real landslides). Table 3.4 shows the thresholds used.

The values differ noticeably between models. XGBoost, for example, flags a landslide at a very low probability of 0.15 to catch as many events as possible, while the stacked model needs a high confidence of 0.90, meaning its combined score separates the two classes very clearly.

3.3.6 Evaluation Setup

All models were evaluated on a held-out test set of 111,000 samples from locations none of the models had seen during training, tuning, or stacking. The models were compared using

Table 3.4: Decision threshold for each model.

Model	Threshold
Random Forest	0.65
XGBoost	0.15
CatBoost	0.75
LightGBM	0.40
Stacked	0.90

accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC. During training itself, 5-fold GroupKFold out-of-fold predictions (also grouped by location) were used to build the stacking meta-learner honestly, so even the meta-learner’s weights were validated out-of-sample.

The final trained system containing the preprocessing pipeline bundled together with the stacked model was saved as a single file "full_pipeline.pkl". It takes raw feature values and directly returns a landslide probability.

4. System Design

4.1 System Architecture

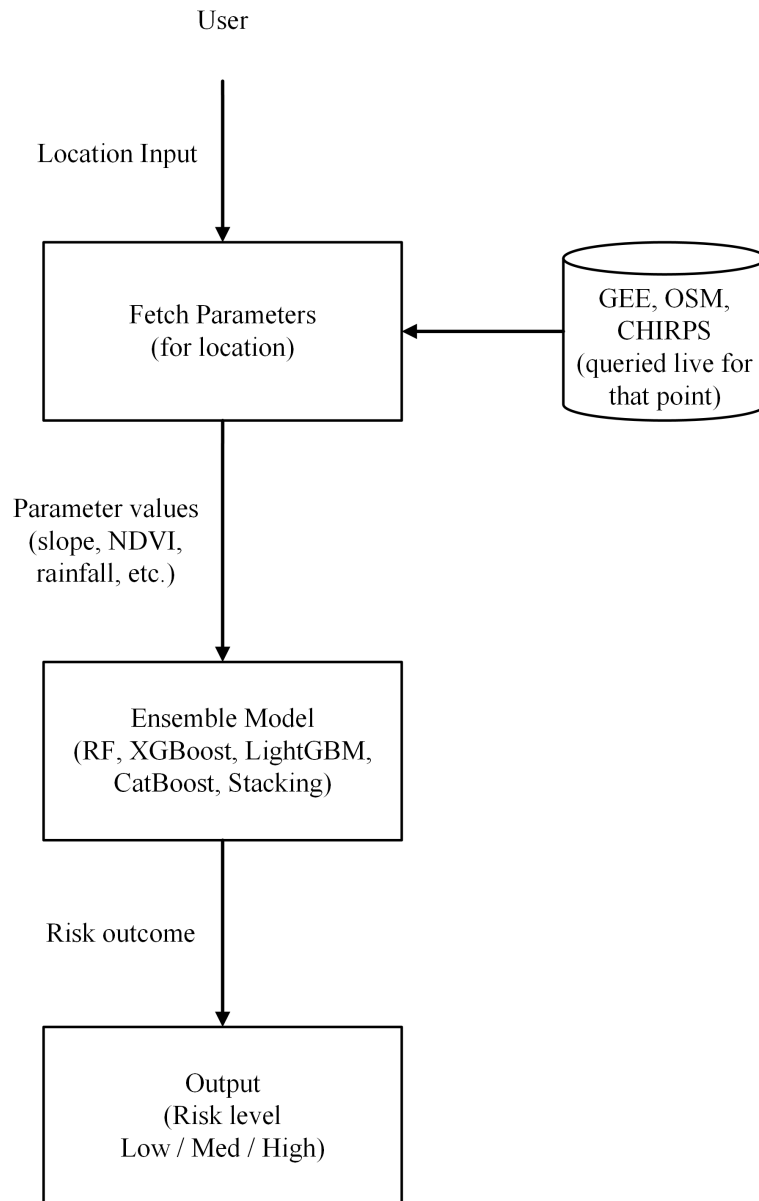


Figure 4.1: System Architecture

Figure 4.1 illustrates the location-specific query flow of the Ensemble Learning Based Landslide Assessment Model, showing how a single user-submitted location is processed into a risk outcome:

- **User Input:** The user submits a location as input to the system, initiating the risk query.
- **Live Parameter Retrieval:** The system fetches the relevant conditioning factor values for the queried location, terrain, vegetation, infrastructure proximity, and rainfall, live from Google Earth Engine, OpenStreetMap, and CHIRPS.
- **Ensemble Model Inference:** The retrieved parameter values are passed to the trained ensemble model, comprising Random Forest, XGBoost, LightGBM, and CatBoost combined through a stacking meta-learner, which classifies the queried location’s landslide risk.
- **Output:** The system returns a risk outcome for the queried location, classified as Low, Moderate, High, or Very High.

4.2 Data Flow Diagrams

The data flow of the system is modeled using Data Flow Diagrams (DFDs) at two levels of detail, presenting the system first as a single high-level process and then decomposing it into its constituent operational modules.

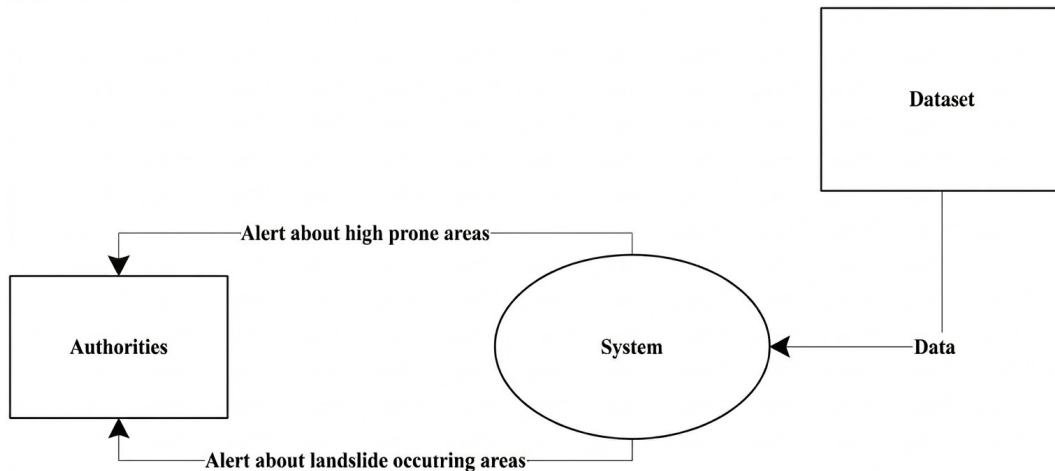


Figure 4.2: Level 0 DFD (Context Diagram)

Figure 4.2 presents the Level 0 DFD (context diagram) of the landslide risk assessment platform. At this stage, the system is represented as a single process interacting with two

external entities: the Dataset, which supplies the conditioning factor and training data, and the User, who receives the landslide risk result for a queried location. This diagram establishes the system’s boundary and its primary inputs and outputs without detailing internal processing logic.

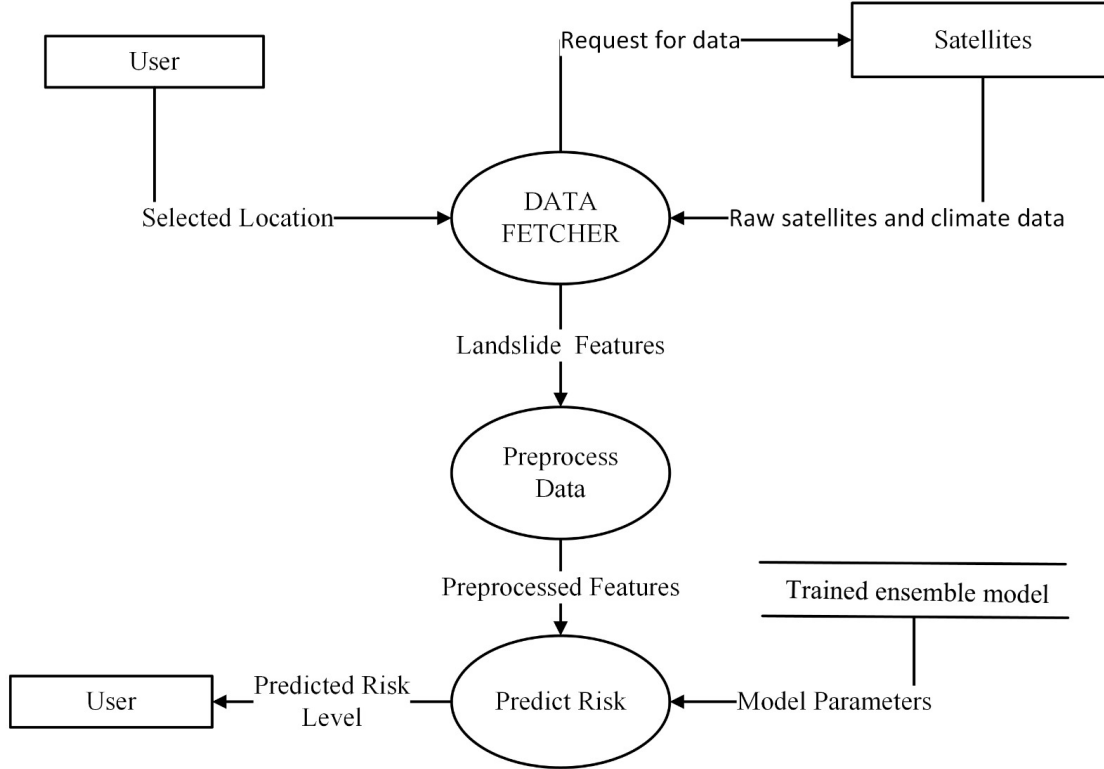


Figure 4.3: Level 1 DFD

Figure 4.3 illustrates how the system processes a landslide susceptibility prediction request. The process begins when the User selects a location, and the corresponding coordinates are sent to Fetch Data, which communicates with external data sources such as GEE, ERA5, and CHIRPS to collect relevant environmental and climatic information, including elevation, slope, NDVI, land cover, and rainfall. The collected raw data are then passed to Preprocess Data, where the features required for prediction, such as slope, aspect, distance to roads and rivers, and rainfall totals, are derived and organized into landslide parameters. These parameters are then forwarded to Predict Risk, which also accesses the Database (Trained Ensemble Model), the only data store shown in the diagram. This database contains the pre-trained model, such as XGBoost, LightGBM, Random Forest, CatBoost, or a stacked combination, developed during the training phase and stored within the system. Unlike the satellite and climate data obtained from external sources, the trained model is maintained by the system for future predictions. Using the processed landslide parameters

and the stored model, Predict Risk calculates the susceptibility probability, applies the classification threshold, determines the corresponding risk level, and returns the final prediction to the User.

4.3 Activity Diagram

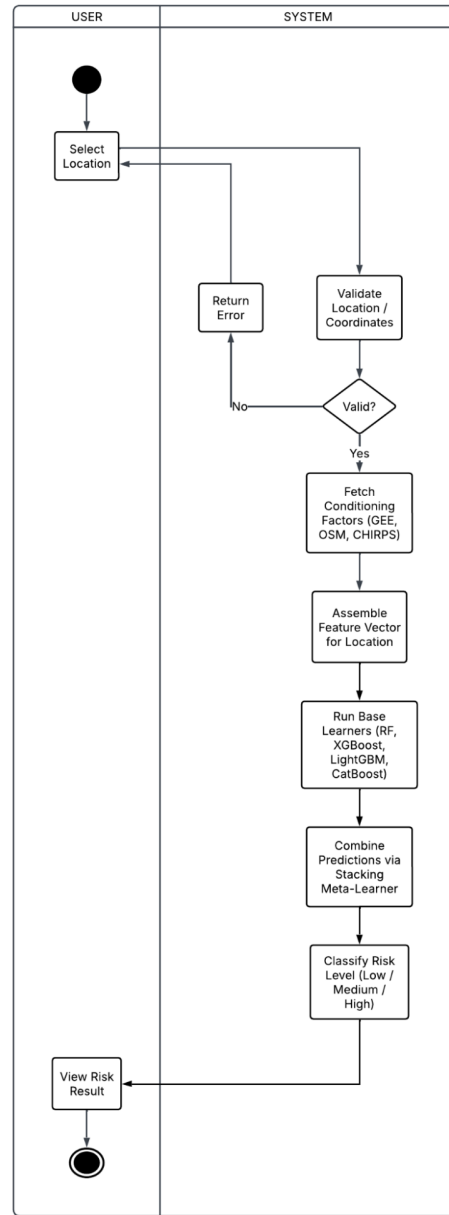


Figure 4.4: Activity diagram of the landslide risk assessment process.

Figure 4.4 presents the activity diagram for the landslide risk assessment system, illustrating the sequence of actions and decision points involved in generating a risk classification for a user-selected location. The diagram is organized into two swimlanes representing the two

participants involved in the process: the User, who initiates the request and receives the final result, and the System, which performs data retrieval, feature assembly, and model inference.

The process begins when the user selects a location. The system first validates the submitted location or coordinates to confirm that they fall within the geographic bounds of the study area and are correctly formatted. If validation fails, the system returns an error to the user, who is prompted to select a location again. This validation loop repeats until a valid location is submitted. After validation, the system gathers location-specific conditioning elements such as terrain, vegetation, infrastructure proximity, and rainfall from Google Earth Engine, OpenStreetMap, and CHIRPS. These factors are then assembled into a feature vector.

This feature vector is passed to the four base learners, Random Forest, XGBoost, LightGBM, and CatBoost, each of which produces an independent prediction. Their outputs are combined through a stacking meta-learner, and the resulting prediction is classified into a risk level of Low, Moderate, High, or Very High. The classified risk level is returned to the user, who views the result, after which the process terminates.

4.4 Deployment Pipeline

The trained system is deployed as a three-tier web application, started with a single command that launches all three services together:

- Model service (FastAPI, port 8000). The trained pipeline "full_pipeline.pkl" is loaded once at startup together with the precomputed per-ward and per-zone feature lookups. For every prediction request the service fills the static features from the lookups, fetches the trailing 30-day accumulated rainfall at the ward or zone centroid from the OpenMeteo weather API, runs the ensemble, and maps the resulting probability to a risk level using the 0.2/0.5/0.9 thresholds. If the weather API is unreachable, a per-month seasonal average is substituted, so the service degrades gracefully instead of failing.
- API backend (Express, port 4000). Serves the public endpoints (wards, zones, statistics, news, alerts) and proxies the prediction endpoints to the model service. User sessions are verified with Clerk, and CORS is restricted to the frontend origin.
- Web frontend (React + Vite, port 5173). Renders the interactive map with the zone grid coloured by risk level, a per-ward risk view, and a click-a-zone prediction interface backed by the model service.

The zone grid used by the map is a precomputed artifact: roughly 1.8 km cells clipped to the district border, each carrying its static feature vector, built by a dedicated script. Since

only rainfall changes between runs, a prediction request is lightweight: one weather lookup and one model inference per queried zone. The conceptual data sources shown in the data flow diagrams are consumed at build time to assemble the feature lookups. The only live external dependency is the rainfall API.

4.5 Web Interface



Figure 4.5: Landing page of the web interface.

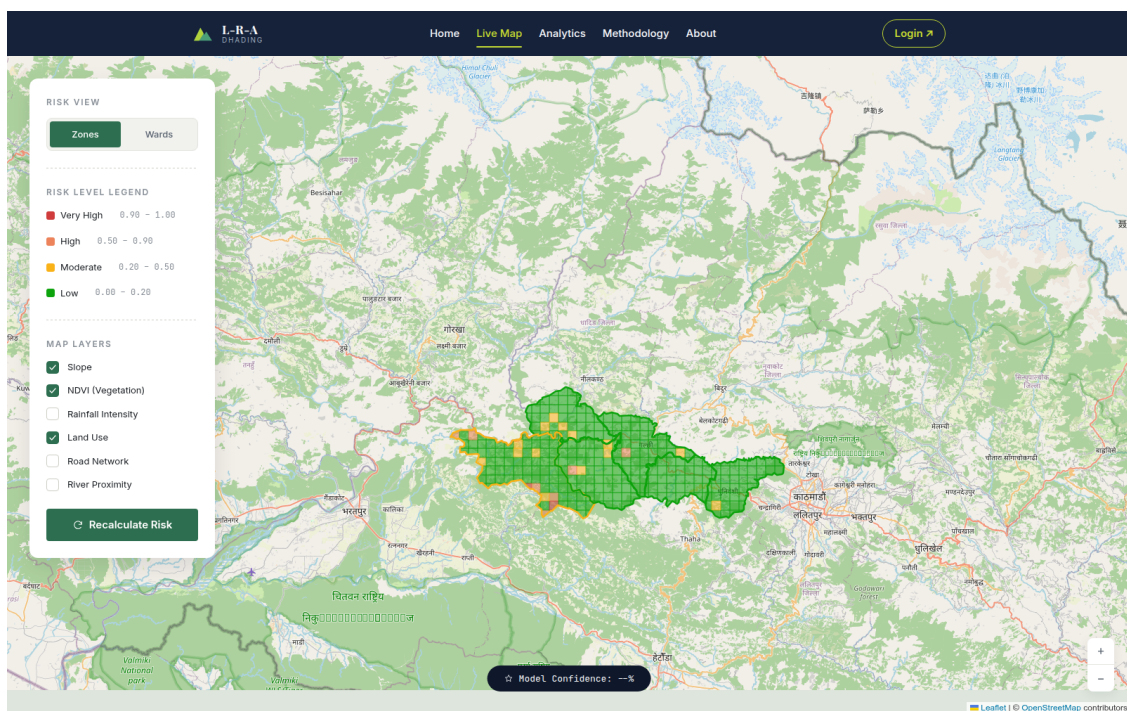


Figure 4.6: Live map showing the zone grid coloured by predicted risk level, with ward boundaries overlaid.

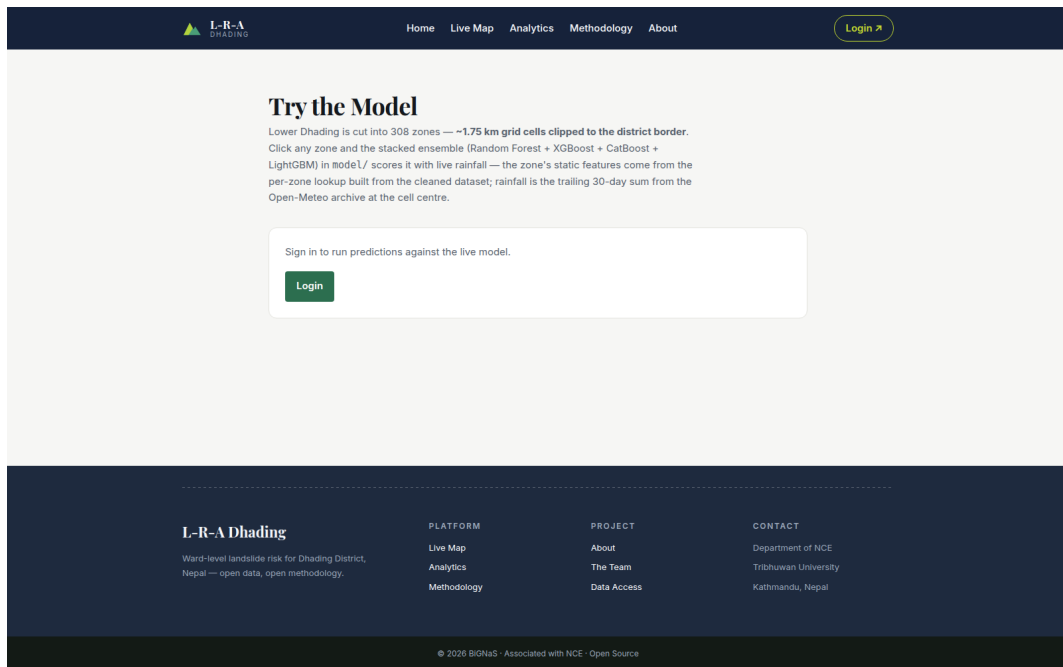


Figure 4.7: Click-a-zone prediction interface.

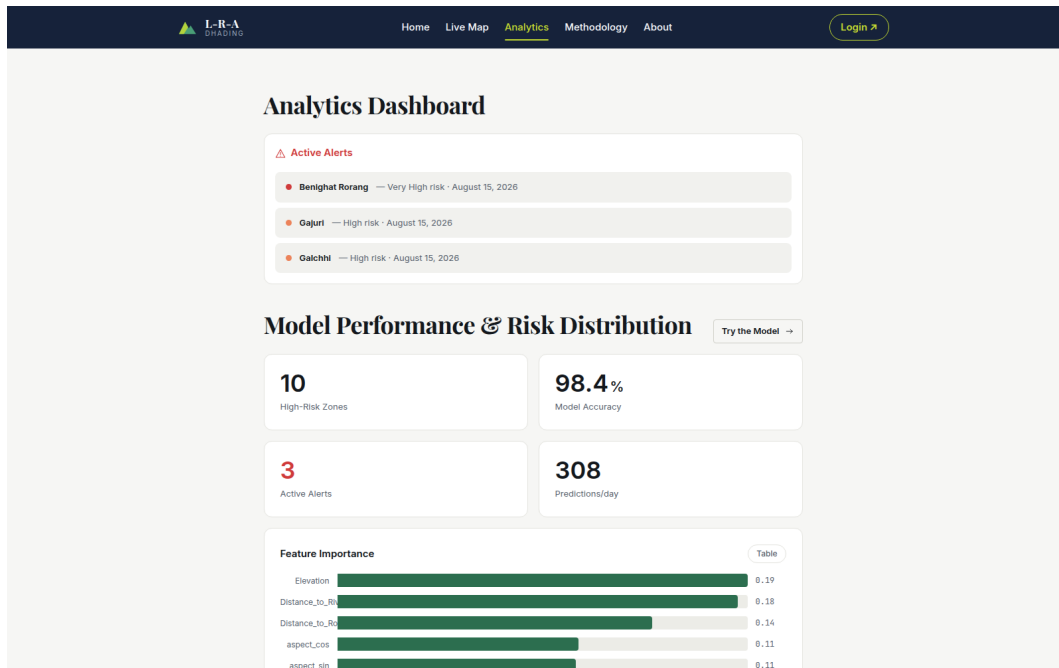


Figure 4.8: Analytics view with model evaluation charts.

The system is delivered through a responsive web interface built with React, providing four main views. The landing page (Figure 4.5) presents the district overview, the latest landslide news feed, and summary statistics for the six municipalities. The live map (Figure 4.6) is the primary risk visualisation: each cell of the zone grid is coloured by its predicted risk level, with ward boundaries drawn as outlines on top, allowing users to identify hazardous localities at a glance. The prediction interface (Figure 4.7) lets a user click any zone to retrieve its conditioning factors and the live rainfall-driven risk assessment for that location. Finally, the analytics view (Figure 4.8) exposes the model evaluation results: ROC and PR curves, the confusion matrix, and the threshold analysis so that the reported performance can be verified directly.

5. Results & Discussion

5.1 Exploratory Data Analysis

We firstly ran exploratory data analysis for understanding the data before building models. The landslide dataset we manually mapped was from April till September for every year. As we are performing the prediction for monsoon, outside of those months we skipped the data.

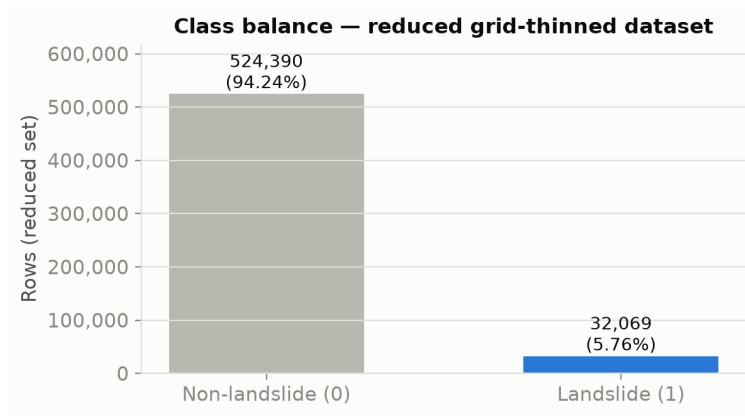


Figure 5.1: Class balance of the raw and reduced training datasets (landslide positives vs. non-landslide samples).

The raw dataset contains 39,807,864 rows of pixel-wise data (about 6.5 GB), extracted at 30 m resolution over Lower Dhading from 2019 to 2025, with a huge extreme class imbalance of roughly 1:1216. We then cleaned and reduced the CSV by retaining all landslide-positive data and excluding negatives by creating a 250 m buffer around each event and selecting only 20 negative candidates from each buffer (spatial buffer exclusion and grid thinning). This reduced the extreme imbalance to about 1:16, which is still imbalanced but far more tractable, as can be seen in Figure 5.1.

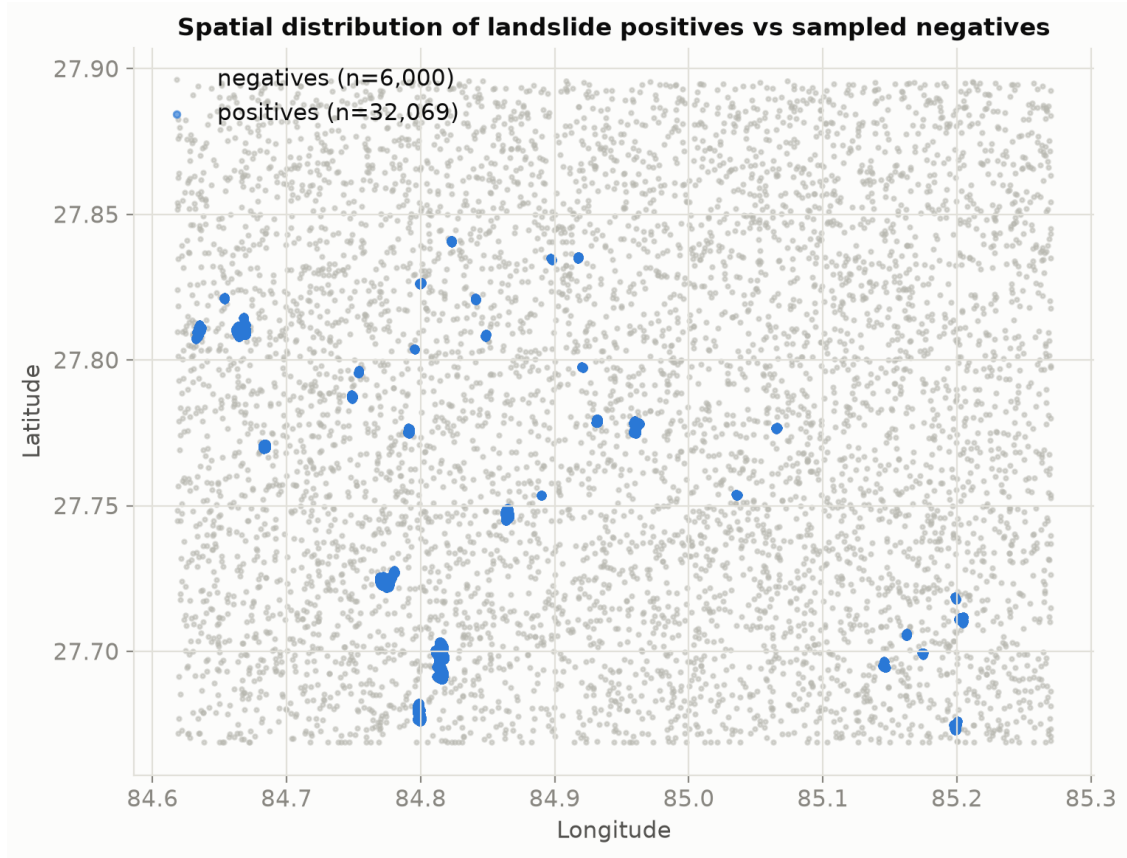


Figure 5.2: Spatial distribution of landslide positives and sampled negatives across the study area.

Figure 5.2 shows the plot of the class labels in a two-dimensional graph between longitude and latitude of a sampled fraction of the dataset .

The steepest slope band ($45\text{-}90^\circ$) has by far the highest landslide-positive rate (Table 5.1).

Table 5.1: Slope distribution of landslide positives.

Slope band	Rows	Positives	Rate
S0 ($0\text{-}15^\circ$)	775k	131	0.017%
S1 ($15\text{-}30^\circ$)	1,998k	1,029	0.052%
S2 ($30\text{-}45^\circ$)	1,090k	1,642	0.151%
S3 ($45\text{-}90^\circ$)	116k	413	0.357%

Low-NDVI (sparsely vegetated) pixels are markedly more prone to landslides (Table 5.2).

Bare and flooded-vegetation pixels show the highest positive rates, while built-up and snow-covered pixels are effectively landslide-free in the inventory (Table 5.3).

Table 5.2: NDVI distribution of landslide positives.

NDVI band	Rows	Positives	Rate
low ($-0.2-0.1$)	35k	324	0.918%
med ($0.1-0.3$)	327k	678	0.207%
high ($0.3-0.5$)	1,001k	917	0.092%
dense_veg (>0.5)	2,612k	1,296	0.050%

Table 5.3: LULC distribution of landslide positives.

LULC name	Rows	Positives	Rate
bare	10k	184	1.82%
flooded_vegetation	1.7k	28	1.67%
water	46k	168	0.37%
shrub_and_scrub	462k	710	0.15%
trees	2,699k	1,942	0.07%
built	411k	48	0.012%
snow_and_ice	3.4k	0	0.00%

5.2 Model Evaluation

Models were evaluated on a spatially held-out test set: the data was split by physical site (latitude and longitude), so no location appears in both training and test data. Per-model decision thresholds were tuned with recall as the priority metric for hazard warning because the ground truth underestimates the true number of landslides. The selected thresholds are 0.65 (Random Forest), 0.15 (XGBoost), 0.75 (CatBoost), 0.40 (LightGBM), and 0.90 (Stacked).

The Stacked ensemble has the highest recall (0.872), greater than its base classifiers LightGBM (0.832), XGBoost (0.852), CatBoost (0.813), and Random Forest (0.670). The Stacked model trades off in the Precision panel (0.841) with LightGBM (0.901) being the highest. Similarly, precision loss leads to a lower F1 score so, Stacked (0.857) lags behind LightGBM (0.865). Since recall is the prioritised metric, we can say that the Stacked ensemble is a better model. (Table 5.4)

Table 5.4: Held-out test metrics per model (spatial split, tuned thresholds).

Model	Acc.	Prec.	Recall	F1	ROC-AUC	PR-AUC	Thresh.
Random Forest	0.969	0.743	0.670	0.705	0.969	0.779	0.65
XGBoost	0.985	0.869	0.852	0.860	0.993	0.935	0.15
CatBoost	0.978	0.803	0.813	0.808	0.989	0.890	0.75
LightGBM	0.985	0.901	0.832	0.865	0.993	0.937	0.40
Stacked	0.984	0.841	0.872	0.857	0.992	0.933	0.90

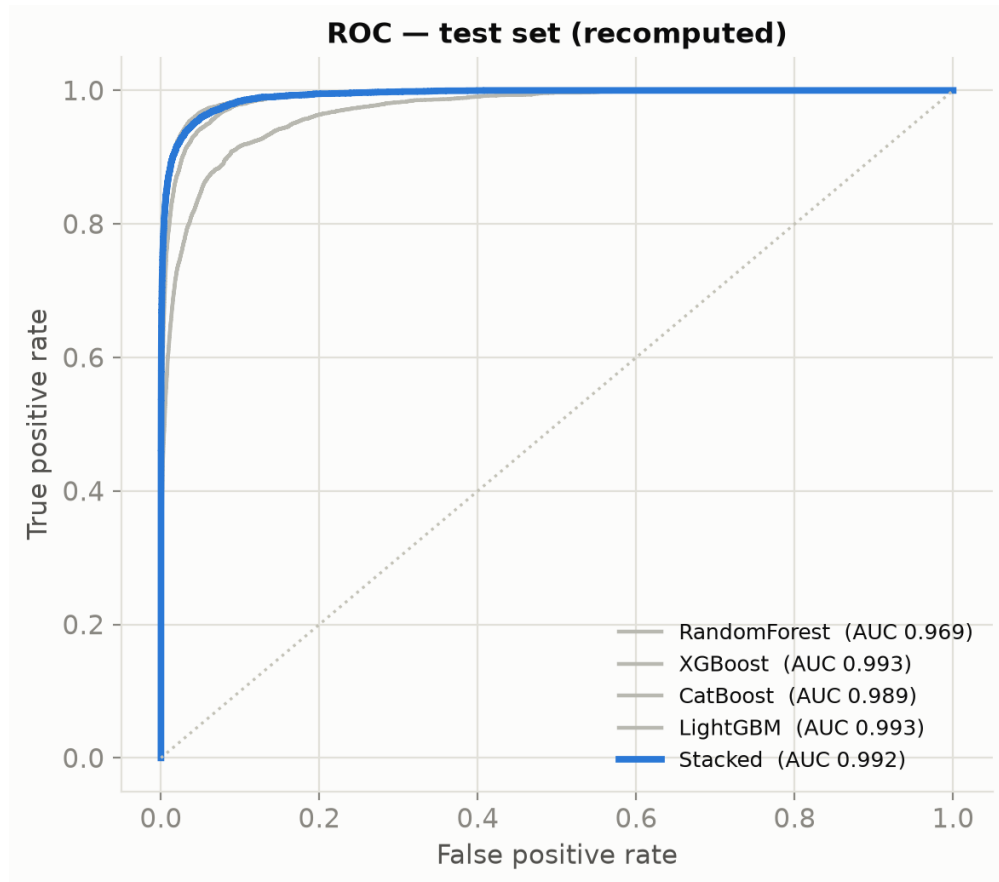


Figure 5.3: ROC curves of the base models and the stacked ensemble on the held-out test set.

All the five models cluster in the top-left corner, very close to 1.0 implying all the models separate landslide vs. non-landslide very well. (Figure 5.3)

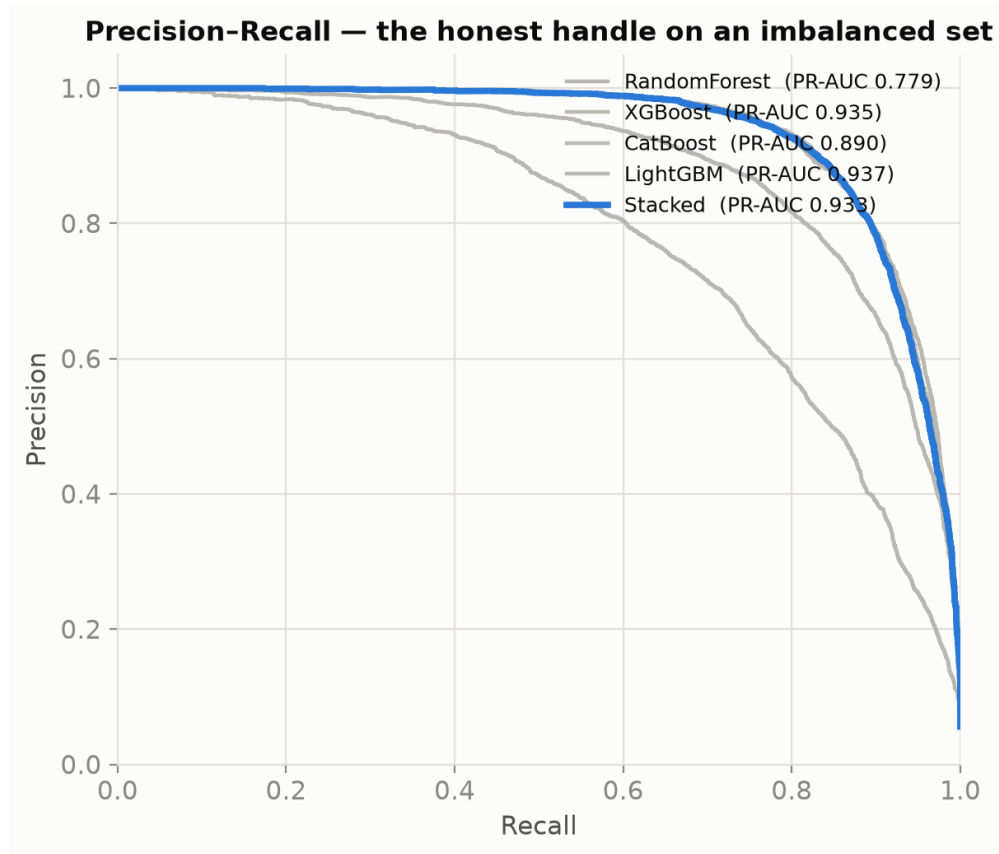


Figure 5.4: Precision-recall curves.

The Stacked ensemble and LightGBM clearly lead the curve implying they keep precision good even while chasing higher recall. Random Forest drops off sharply suggesting that the model is overconfident. (Figure 5.4)

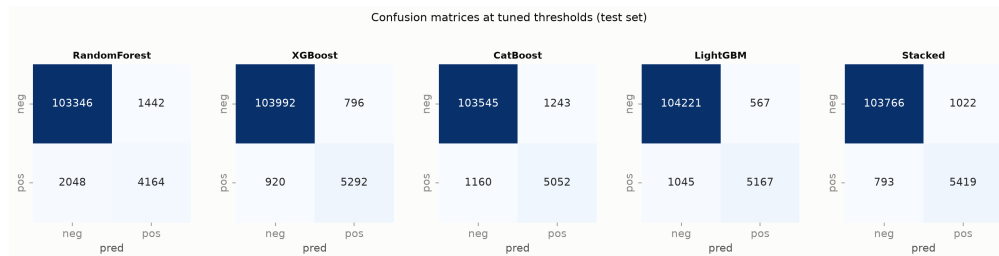


Figure 5.5: Confusion matrices of all five models.

The confusion matrices of each of the 5 models are shown by Figure 5.5 showing the various values of the false negatives and the false positives that significantly affect the model's recall and precision. The Stacked ensemble evaluated with 5,419 true positives and only 793 missed landslides which results in the 0.872 recall.

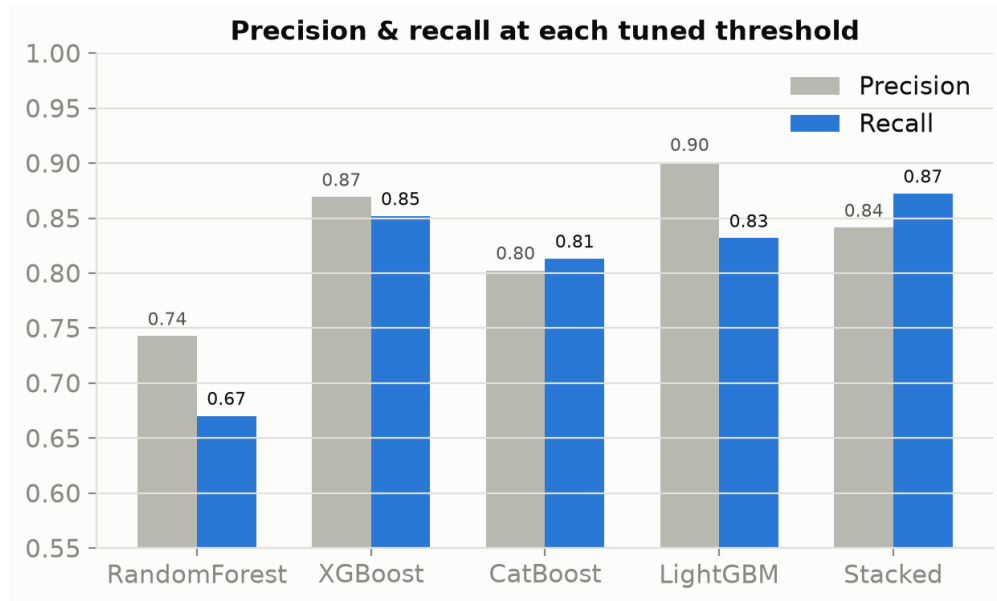


Figure 5.6: Effect of the decision threshold on precision and recall.

Each model has its own tuned threshold to balance precision and recall best for hazard warning. The thresholds were obtained by rigorously running the models using Optuna for tuning. The thresholds were found to be 0.65 for Random Forest, 0.15 for XGBoost, 0.75 for CatBoost, 0.40 for LightGBM and 0.90 for the Stacked ensemble. The precision and recall obtained are shown in Figure 5.6. The stacked model is itself a small model that takes the four base models' scores and combines them.

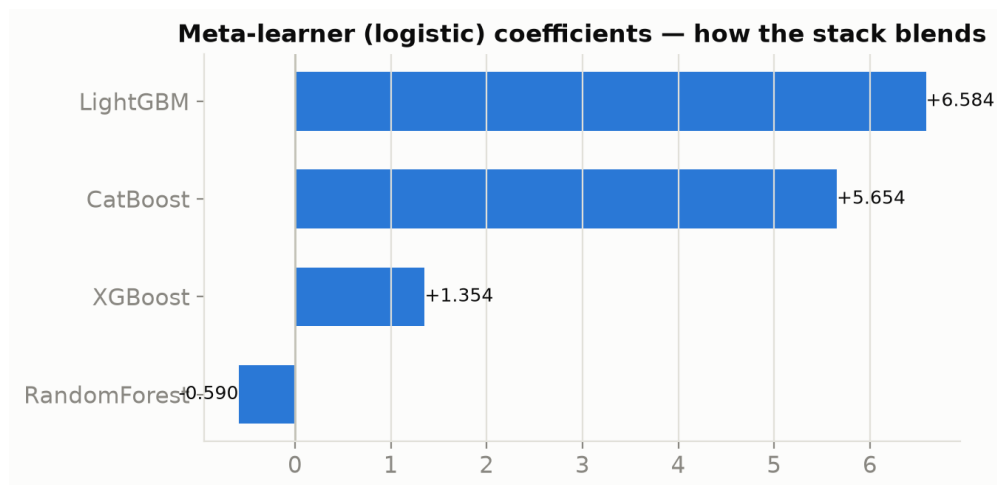


Figure 5.7: Learned weights of the meta-learner over the base models.

Figure 5.7 shows the learned weight each base model was given.

All four base learners exceeded an ROC-AUC of 0.96 on the spatially held-out test set, with LightGBM the strongest individual learner (ROC-AUC 0.9928, PR-AUC 0.937). The

stacked ensemble achieved the highest recall of any configuration (0.872) at its 0.90 threshold while matching the best base learner on ranking quality (ROC-AUC 0.9924) so the stacked model is the configuration deployed in the live system.

5.3 Susceptibility Zonation

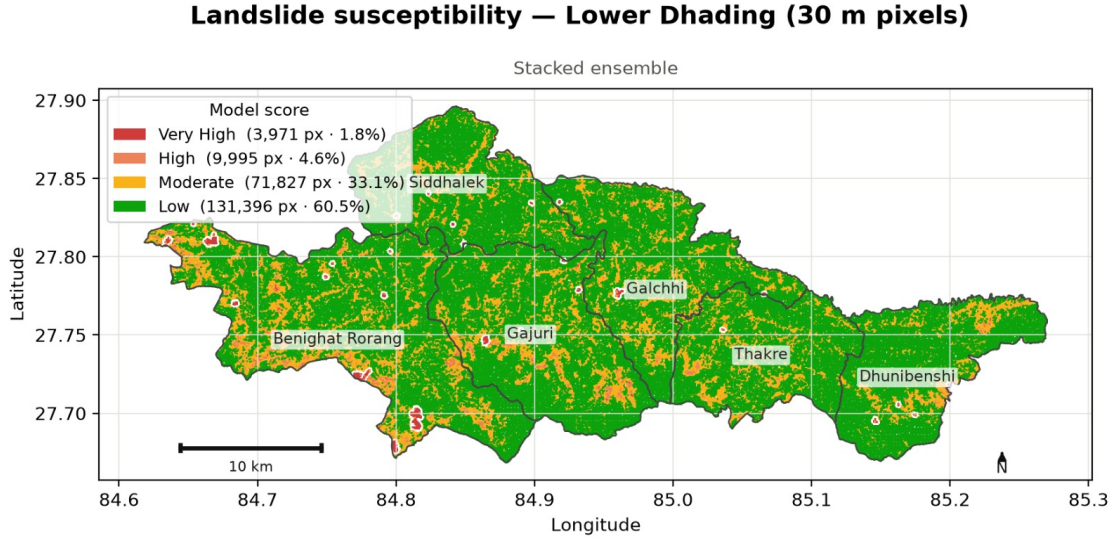


Figure 5.8: Landslide susceptibility zonation map of Lower Dhading.

The trained stacked ensemble was used to score the entire study area, producing the susceptibility zonation map shown in Figure 5.8. Of the 308 zones covering Lower Dhading, 283 (91.9%) fall into the Low band, 15 (4.9%) into the Moderate band, 8 (2.6%) into the High band, and 2 (0.6%) into the Very High band. The ten zones rated High or Very High are concentrated in Benighat Rorang municipality, which also carries the highest ward-level score of the six municipalities (0.269), followed by Galchhi (0.066), Gajuri (0.063), Siddhalek (0.044), Dhunibenshi (0.029), and Thakre (0.010). The highest-risk areas lie along the Prithvi Highway corridor and the steep southern slopes of the district, where road-cut excavation and heavy monsoon rainfall coincide with the highest predicted probabilities.

5.4 Discussion

The results of the model are consistent with the report by Kalantar et al. [8] that a well-constructed ensemble outperformed every individual classifier on the same evaluation.

Recall was prioritized because the inventory contained only 33 manually mapped events. Any other landslides outside these events were treated as negatives which surely increased the false negatives. So, recall is the better measure for how many landslide pixels the model flagged. The stacked model’s recall of 0.872 matters more than accuracy of 0.984 and

precision of 0.841. This is generally true for hazard-warning use cases, where missing real events can be dangerous than false warnings.

The EDA findings reinforce the physical plausibility of the model. Landslide-positive rates rise sharply with slope (0.357% on 45-90° slopes vs. 0.017% on gentle terrain), collapse as NDVI increases (0.918% on low-NDVI pixels vs. 0.050% on dense vegetation), and are highest on bare and flooded-vegetation land cover. The model was consistent with the conditioning factors (slope, vegetation, land cover) identified as dominant drivers across the wider Nepal Himalaya [7, 13]. The out-of-fold logistic-regression meta-learner (Figure 5.7) learned how to combine the four base probabilities, and the stacked model’s recall gain over every base learner confirms that the four models are complementary rather than redundant.

Previous work associated with this district has been either field-based inventory documentation [5] or a single-classifier, single-municipality assessment using classical multi-criteria analysis [3]. The study extended both by covering all six municipalities of Lower Dhading with a supervised ensemble trained on pixel-level, monthly-resolved data and it delivering the result as an operational query system rather than a static map. The study addressed the inventory and modelling gaps documented at the national scale [1, 6].

The deployed system maps the model’s continuous score onto four risk levels using thresholds of 0.2, 0.5, and 0.9, and surfaces them per zone of a fine grid clipped to the district border. The study provides a location-specific screening tool before and during the monsoon season for municipal authorities.

6. Conclusion

The study intended to build a landslide susceptibility assessment system for Lower Dhading, Nepal. The objectives set for the study: building an ensemble machine learning model trained and validated on multi-source geospatial factors and generating susceptibility outputs through a querying system were achieved.

The dataset assembled for the study is one of the first pixel level monthly-resolved landslide susceptibility dataset covering the six municipalities of Lower Dhading. The dataset was reduced to a 556,459 row monsoon-season training set through spatial buffer exclusion and grid thinning from a 39.8 million rows dataset at 30 m resolution.

Methodologically, all four base learners exceeded 0.96 ROC-AUC on the spatially held-out test set, confirming that the conditioning factors carry strong predictive signal. The stacked ensemble achieved the highest recall of any configuration (0.872) at its tuned threshold of 0.90 while matching the best base learner on ranking quality (ROC-AUC 0.9924). This demonstrates that a well-constructed stacking ensemble can outperform its best constituent for the hazard-warning objective.

The project delivered a working end-to-end system. The trained pipeline fills static features from pre-computed lookups and a live 30-day rainfall from a free weather API. The model is made available at a web interface that returns four-level risk assessments for any queried location. The system also lists out recent landslides that have occurred in the Area of Interest through BIPAD’s API. The system also provides disaster management agencies, local authorities and communities with location-specific decision support and planning prospects for the monsoon-season for the district.

In total, the project demonstrates that localized, data-driven susceptibility assessment is feasible with freely available geospatial data and ensemble learning. The project can also be used as a practical tool to support disaster preparedness in vulnerable municipalities.

7. Limitations and Future Work

The research delivers a working model for susceptibility assessment of Lower Dhading but many methodological and operational limitations were encountered.

Reduction of the data from 39.8 million rows to 556 thousand rows was done through buffer exclusion and grid thinning to make training possible. This means that a vast majority of the unlabeled pixel pool was discarded. A semi-supervised approach would leverage the unlabeled data together with the small labeled inventory resulting in improved generalization without any information loss. Any probable negatives or positives near mapped events are excluded with the use of a 250 m buffer. This could lead to a loss of informative pixels that might distinguish area near landslides from area far from landslides. The study will benefit from using the buffer as a third class, neither negative nor positive.

All the conditioning factors of the model rely on derived and modeled data from various satellite sources. This introduces resolution and revisit times constraints to the model. The model will perform greater if low-cost sensors such as rain gauges, soil moisture probes and slope movement monitors are deployed at the identified high-risk zones as they provide ground truth for validation, more reliable rainfall estimates, and earlier detection of slope instability than remote sensing alone. The system currently only computes and displays risk on demand, but does not send out alerts when there is a higher risk. An additional alerting layer that notifies disaster focal points whenever any zone reaches the very high risk band. Also, risk is currently recomputed per query because rainfall is the only time-varying feature. Scheduling the model to re-run automatically at fixed intervals would keep the published risk layer updated without any user action.

Finally, the model was trained and validated only on Lower Dhading's six municipalities. Expanding the study area to the rest of Dhading and then to neighboring mid-hill districts that share similar geology and road-expansion pressures would test how well the learned relationships transfer spatially. it would also give more municipalities the same tool.

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